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Enhancing biodegradability of industrial park wastewater by packing carriers and limited aeration in the hydrolysis process

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ABSTRACT

This study aims to ameliorate effects of packing carriers, filling ratio, temperature and limited aeration on hydrolysis performance in a chemical industrial park wastewater treatment plant. *Illumina-MiSeq* sequencing was applied to compare microbial community structure between biofilm attached on carriers and activated sludge in a full-scale hydrolysis tank. Packing carriers intensified the plentifulness of microbial community, enriched hydrolytic bacteria (*Betaproteobacteria*, *Gammaproteobacteria*, *Sphingobacteria*, *Clostridia*, *Bacteroidia*), and enhanced biodegradability of organics in the wastewater. Rising carriers filling ratio increased first-order kinetic constant of hydrolysis from 0.015 to 0.080 h^{-1}, and decreased fractions inert to hydrolysis. Rising temperature stimulated enzymatic activity for hydrolysis with activation energy of 80.22 kJ/mol. Limited aeration further enriched hydrolytic bacteria, inhibited sulfate-reducing bacteria and intensified biodegradability of organic substances. Variations of fluorescent organic substances and ammonia nitrogen release during hydrolysis confirmed that hydrolysis was strengthened by packing carriers, increasing filling ratio, enhancing temperature and introducing limited aeration. Packing carriers and limited aeration were techno-economic methods for enhancing hydrolysis performance in the chemical industrial park wastewater treatment plant by enriching hydrolytic bacteria and stimulating their activities.

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1. Introduction

There are more than 2500 provincial-level industrial parks in China, which contribute more than 50% of the national gross domestic product (Hu et al., 2019a) but discharge enormous industrial wastewater simultaneously (Hu et al., 2019b) with characteristics of omnigenous compositions and high concentration of pollutants (Zhu et al., 2016). Moreover, these compositions are usually toxic and non-biodegradable (Liu et al., 2019), and probable cannot be entirely degraded (Ma et al., 2016). Several treatment methods based on physical, chemical, physical-chemical, and biological mechanism were used widely in the chemical industrial park wastewater treatment plant (WWTP) (Hu et al., 2019b). The low-cost and energy-saving biological treatment technology was highly appreciated among them. Nevertheless, the performance of removing refractory compounds with biological treatment is not efficient, resulting in the low organic compounds removal efficiency, poor carbon source for denitrification and low utilization ratio of energy. Ultimately, the concentrations of chemical oxygen demand (COD) and total nitrogen (TN) could not meet requirements in "Discharge Standard of Pollutants for Municipal Wastewater Treatment Plant of China (GB18918-2002)". Therefore, it is indispensable to develop the treatment technology for certain degradation-resistant organic pollutants to enhance their removal efficiencies.

Hydrolysis process is often employed to enhance biodegradability of the industrial park wastewater, making it conducive to the

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succedent biological treatment. However, the composition of industrial park wastewater is very complex, making conventional hydrolysis by suspended sludge under anaerobic conditions certainly difficult (Wu et al., 2015; Zhang et al., 2016). Intensive efforts have been conducted in recent years to enhance hydrolysis efficiency, and are very helpful to give some enlightenment for industrial park wastewater treatment. In some cases, carriers could store up high concentrations of active biomass and enrich the growth of hydrolytic bacteria, which are indispensable for the recalcitrant organic compounds degradation (Gu et al., 2014; Huang et al., 2017; Zheng et al., 2019). Micro-aeration by introducing small amounts of oxygen into a bioreactor to stimulate biological activities of both anaerobic and aerobic biomass (Botheju and Bakke, 2011; Kong et al., 2019; Zhou et al., 2014), has the potential to accelerate hydrolysis. Wu et al. (2015) reported that the biochemical oxygen demand (BOD_5)/COD of the petrochemical wastewater increased from 0.23 to 0.43 by limited aeration in the hydrolysis tank. However, there was limited information on effects of carrier and limited aeration on hydrolysis performance of industrial park wastewater treatment. More detailed research is needed for the combination of limited aeration and carriers in hydrolysis tanks, which is a potentially effective strategy to enhance biodegradability of industrial park wastewater.

In this study, carriers were firstly suspended in the hydrolysis tank of an industrial park WWTP to incubate biofilm attached on their surfaces, and then used to investigate effects of filling ratio, temperature and limited aeration on hydrolysis efficiency and biodegradation of treated wastewater. *Illumina-MiSeq* sequencing was applied to investigate the microbial community structure and population composition of biofilm and activated sludge (AS) in the hydrolysis tank. The hydrolysis process was also evaluated by a first-order kinetic model. The dissolved organic matters (DOM) in the industrial park wastewater were also characterized by the three-dimensional excitation-emission matrix (3D-EEM) fluorescence spectroscopy. The results of this study are expected to provide some feasible enhancement strategies on hydrolysis of industrial park wastewater.

2. Materials and methods

2.1. Activated sludge

AS used in this study was obtained from hydrolysis tank of a chemical industrial park WWTP (Jiangsu, China) operated stable for two years with wastewater flow rate of 20,000 m^3/d treated by hydrolysis – anoxic/anaerobic/oxic (AAO) - sequencing batch reactor (SBR) process. The hydraulic retention times (HRT) of hydrolysis, anoxic, anaerobic, oxic and SBR were 24, 4, 4, 12 and 4 h. The wastewater treated by the chemical industrial park WWTP was a mixed wastewater, including domestic wastewater, pharmaceutical industry wastewater, petroleum industrial wastewater, textile dye industry wastewater, etc. The main characteristics of samples from the WWTP were as follows: pH of 7.5–8.5, soluble COD (SCOD) of 112.8–158.2 mg/L, BOD_5 of 6.8–16.2 mg/L, NH_4^+-N of 8.5–11.0 mg/L, and TN of 29.9–40.5 mg/L.

2.2. Carriers for biofilm attachment

The LEVAPOR biocarriers (LEVAPOR GmbH Biofilm Technologies, Leverkusen, Germany), which were made from polyethylene (PE) with total surface area of 20,000 m^2/m^3, were employed as carriers for biofilm attachment in this study. The biofilm attached on PE carriers was grown and incubated for three months in the hydrolysis tank of the WWTP before conducting the batch tests. Part of the PE carriers incubated for three months were taken out and placed in a lab-scale hydrolysis reactor under the condition of limited aeration for another 30 d, and the industrial park wastewater was fed into the reactor to maintain HRT of 24 h.

2.3. Experimental procedure

In order to investigate effects of carrier filling ratio, temperature and limited aeration on hydrolysis efficiency, some lab-scale reactors of 5 L equipped with immersion thermostat and magnetic stirrer were designed. All reactors were initially seeded with the same AS and carriers with biofilm attached on their surfaces, and then fed with influent of the WWTP to maintain COD concentration at approximately 125–170 mg/L. The stirring rate was maintained at 120 rpm to make AS and carriers mix homogenously. After start-up, samples were taken from each reactor using a pipette at 0, 3, 9, 18, 28, 38 and 48 h for the immediately analysis of COD, BOD_5 and NH_4^+-N concentrations in duplicate after filtration by 0.45 μm vinyl cellulose membranes. The fluorescence characterization of DOM collected at 0, 9, 28, and 48 h were also analyzed using the 3D-EEM fluorescence spectroscopy.

The operational conditions are shown in Table 1. Firstly, the effect of carrier filling ratio on hydrolysis was concurrently investigated at three carrier filling ratios of 0, 30%, and 60% under anaerobic condition. And the temperature was sustained at values of 25 ± 0.1 °C. Secondly, under anaerobic condition, the batch tests of hydrolysis were also conducted at 15 ± 0.1, 25 ± 0.1 and 35 ± 0.1 °C to investigate temperature effects at carrier filling ratio of 30%. Lastly, two reactors were also operated in parallel under anaerobic condition and the condition of limited aeration with average dissolved oxygen (DO) of 0.2–0.5 mg/L. The carrier filling ratio and temperature were controlled at 30% and 25 ± 0.1 °C. All the experiments were conducted in duplicate.

2.4. Microbial diversity analysis

Microbial community structure of an AS sample (from the full-scale hydrolysis tank) and two biofilm samples (from carriers suspended in the hydrolysis tank for three months, and the hydrolysis reactor under limited aeration for 30 days) were analyzed by *MiSeq* sequencing to compare microbial community structure evolution. The detailed experimental steps of DNA extraction, PCR amplification, 16S rRNA gene sequencing, biodiversity analysis and phylogenetic classification were all in accordance with the reported methods described by An et al. (2017). Finally, sequencing produced more than 34,000 high-quality V3–V4 tags for 16S rRNA genes with an average length of 436 bp by setting a 0.03 distance limit (α). To equitably compare the two samples at the same sequencing depth, the sub.sample command of Mothur program was employed for normalization, and 34,094 sequences were extracted from each dataset. Based on cluster information, parameters referring to species richness (Ace and Chao), community diversity (Shannon and Simpson) and sequencing depth (Good's Coverage) were calculated by Mothur software.

Table 1
Experimental parameters for batch tests to investigate effects of carriers filling ratio, temperature and limited aeration on hydrolysis performance.

Parameters	Test 1	Test 2	Test 3
Carrier filling ratio (%)	0, 30 and 60	30	30
Temperature (°C)	25	15, 25 and 35	25
Limited aeration (mg DO/L)	None	None	0.2–0.5

2.5. Kinetics model of COD hydrolysis

Because some organic substances in chemical industrial wastewater are non-biodegradable (Kong et al., 2019; Zhu et al., 2016), the kinetics of COD hydrolysis follows a first-order kinetic equation shown in Eq. (1)

$$dCOD / dt = -k(COD - COD_I) \quad (1)$$

where COD and COD_I is the total COD and inert COD in the reactor, mg/L; k is the first-order kinetic constant, h^{-1}; t is the hydraulic retention time, h. Integration of Eq. (1) at boundary conditions of ($t = 0$, $COD = COD_0 + COD_I$) gives,

$$COD = COD_0 e^{-kt} + COD_I \quad (2)$$

where COD_0 is the initial biodegradable COD in the reactor, mg/L. The activation energy (Ea, J/mol) of hydrolysis process with carriers for COD removal from industrial park wastewater is calculated using the Arrhenius equation in Eq. (3) with k values determined at different temperatures (T, K).

$$k = Ae^{-\frac{E_a}{RT}} \quad (3)$$

where, A is the pre-exponential factor, h^{-1}; R is the gas constant, 8.3145 J/(mol·K).

2.6. Other item analysis and statistical analysis

Concentrations of COD, BOD_5 and NH_4^+-N in samples were analyzed according to standard methods (Chinese NEPA, 2012). DO concentration in the reactor was tested by an HQd30 m (HACH, USA). The 3D-EEM fluorescence spectra of the supernatant were conducted using a fluorospectrophotometer (RF-5301pc, Shimadzu, Japan). The parallel factor (PARAFAC) analysis was performed for all EEM spectra in MATLAB R2013a (Mathworks, USA) using the DOMFluor.

Correlation coefficient (r) between released NH_4^+-N and decreased COD were conducted by using SPSS 19.0 software.

3. Results and discussion

3.1. Microbial community

The integrated investigation of microbial community composition and diversity of AS and biofilm attached on the surface of carriers was performed through *MiSeq* sequencing before conducting the batch tests. The purposes of this section were to show microbial evolution of packing carriers and limited aeration in the hydrolysis tank for industrial park wastewater treatment, and to elucidate the composition of microorganisms involved in the following batch tests.

3.1.1. Richness and diversity of bacteria phylotypes

The statistical results of diversity analysis (Table 2) showed that the microbial community structure and composition in AS differed from biofilm, although the microorganisms in biofilm originated from AS in the hydrolysis tank. It is conjectured that microbial growth may be different in the hydrolysis tank and on the surface of PE carriers. By performing the alignment at an approximate 436 bp length, 1077 (AS) and 1142 (biofilm) operational taxonomic units (OTUs) were clustered at a 0.03 cutoff level. The total OTUs estimated through Chao's estimation with infinite sampling was 1112 (AS) and 1186 (biofilm), indicating that biofilm had higher microbial richness than AS (Cheng et al., 2017). Higher Shannon value and lower Simpson value demonstrated that microbial community of biofilm was more diverse in terms of distribution and evenness than that of AS (Cheng et al., 2018; Choi et al., 2017). Moreover, the higher Shannon index suggested that the enriched OTUs in biofilm community were distributed more evenly than that in AS (Ma et al., 2013a). All four indices in Table 2 illustrated that biofilm had higher diversity than AS (Matar et al., 2017), indicating that PE carriers provided better substrates for hydrolysis by disintegrating the floc structure of AS. The Good's coverage obtained for each sample was both above 0.99, revealed that the sequences generated at this sequencing depth could well represent the bacterial communities of AS and biofilm samples.

3.1.2. Bacterial taxonomic identification

To identify the changes of bacterial community structure and composition between AS and biofilm, the phylogenetic spectrums were employed to distinguish taxa levels from phylum to genus (Fig. 1). Fig. 1a depicts the relative bacterial abundance on the phylum taxonomic level. 42 bacterial phyla, demonstrating greatly high diversity, were detected in microbial communities of the two samples. The majority of the sequence reads were assigned to six major phyla: *Chloroflexi*, *Proteobacteria*, *Acidobacteria*, *Bacteroidetes*, *Chlorobi* and *Gemmatimonadetes*, which in total contributed to 80.8% (AS) and 84.8% (biofilm) of the entire sequence reads, respectively. Previous studies reported that these phyla were ubiquitous in AS and biofilms (Choi et al., 2017; Ma et al., 2013a; Xie et al., 2014). The dominant microorganisms in biofilm resembled that of AS, indicating that the microorganisms derived from AS might induce an effect of colonization on the surface of PE carriers. However, the differentiation of percentages procured for each phylum sample was also observed. For example, *Chloroflexi*, playing a role in degrading carbohydrates and cellular materials (Choi et al., 2017), was the most dominant phylum in AS, accounting for 35.8% of the total sequence reads. While *Proteobacteria* replaced *Chloroflexi*, and became the most dominant phylum in biofilm (35.7%), indicating that PE carriers promoted its growth. The total sequence reads of *Bacteroidetes* in the biofilm (8.4%) were higher than to those in the AS (5.6%). In the anaerobic condition, *Proteobacteria* could induce cell lysis and release intracellular substances, and then *Bacteroidetes* utilized the secondary substrate for hydrolytic fermentation to proliferate (Han et al., 2015; Kim et al., 2009). Besides, *Firmicutes* could be related to denitrification and complex organic matters degradation (Kumar et al., 2010). The relative abundance of *Firmicutes*, in particular *Clostridium* sp., having spores to stand up to the unfavorable environment (Xie et al., 2014), was 1.0% in biofilm, higher than that in AS (0.4%). After that, additional research down to the class and genus level was performed to recognize bacterial pioneers between AS and biofilm.

Table 2
Richness and diversity estimators of microbial communities of activated sludge, and biofilm under anaerobic and limited aeration conditions ($\alpha = 0.03$).

Sample	Reads	Normalization	OTUs	Ace	Chao	Shannon	Simpson	Coverage (%)
AS	40,414	34,094	1077	1114	1112	5.49	0.0191	99.8
Anaerobic biofilm	34,094	34,094	1142	1176	1186	5.70	0.0105	99.7
Limited aeration biofilm	45,053	34,094	1163	1191	1193	5.75	0.0116	99.8

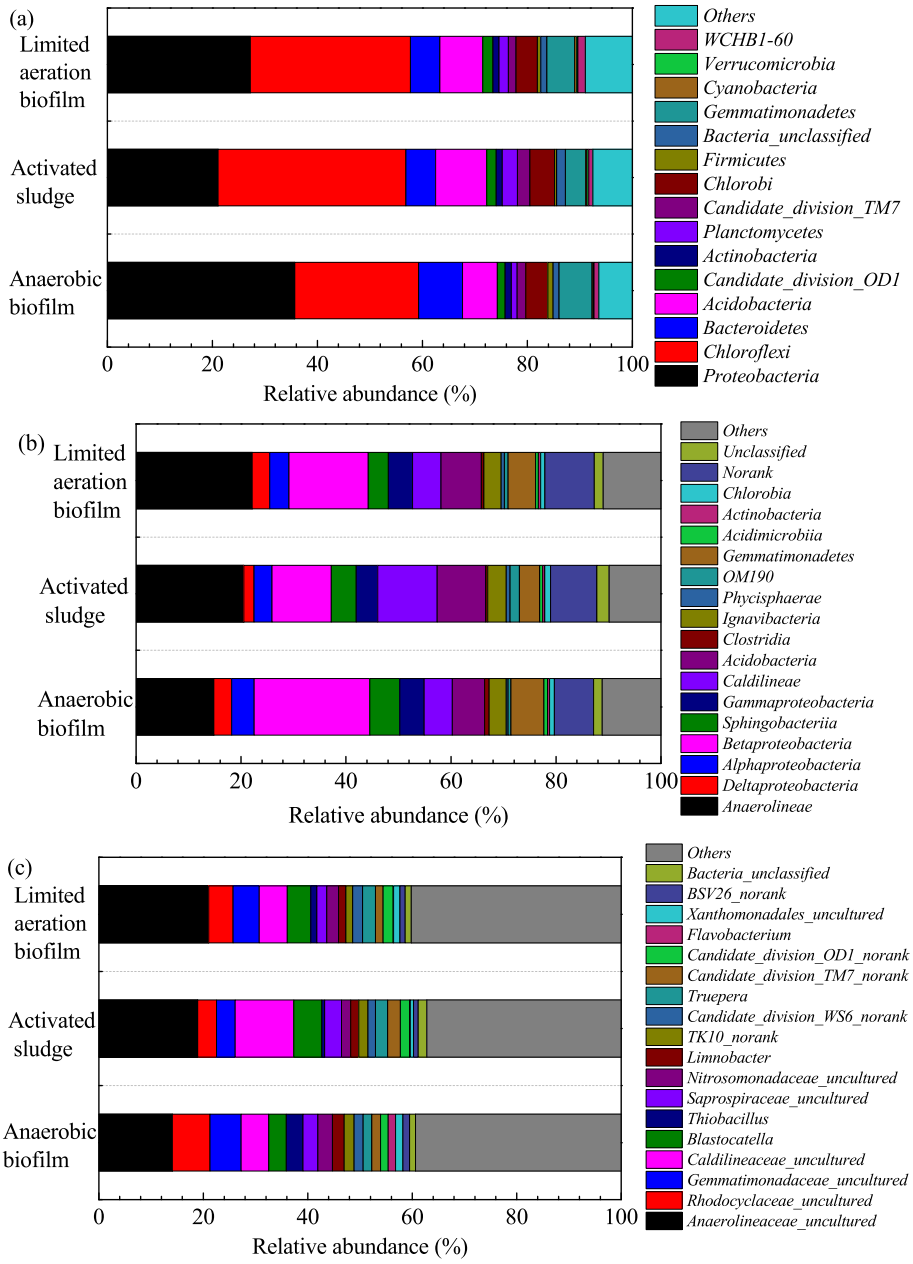


Fig. 1. The relative abundances of the predominant phylogenetic groups of activated sludge, biofilm under anaerobic and limited aeration conditions at the (a) phylum, (b) class and (c) genus level. Classes and genera making up less than 1.0% of the total sequences in each sample were classified as others.

Fig. 1b illustrates the class taxonomic level identification of the microbial communities. 89 bacterial classes were detected in these two communities. However, the majority of sequences pertained to 18 classes (i.e., >1% relative abundance), which in total accounted for 88.3% and 88.6% of the total phyla in AS and biofilm, respectively. Compared to those in AS, *Betaproteobacteria* (*Proteobacteria*), *Anaerolineae* (*Chloroflexi*), *Gemmatimonadetes* (*Gemmatimonadetes*), *Acidobacteria* (*Acidobacteria*), and *Sphingobacteriia* (*Bacteroidetes*) were the dominating subgroups of the microbial community in the biofilm. *Anaerolineae* of phylum *Chloroflexi*, activated in the anaerobic biogas production processes (Wu et al., 2016), was the most dominant class in AS (20.5%), whereas its abundance decreased and became the second dominant class in biofilm (14.8%). On the contrary, the proportions of *Beta-*, *Gamma-*, *Alpha-* and *Deltaproteobacteria* affiliated to phylum *Proteobacteria*, which were responsible for polysaccharides utilization or butyric acid fermentation (Niu et al., 2016), were all obviously higher in biofilm than those in AS. The relative abundance of *Betaproteobacteria* was 11.3% in AS, but increased significantly to 22.0% in biofilm. With this change, *Betaproteobacteria* became the most dominant class in biofilm. Similarly, the relative abundance of *Gemmatimonadetes* and *Sphingobacteriia*, accounted for 3.9% and 4.7% of total sequences in AS, and increased to 6.3% and 5.7% in biofilm. *Sphingobacteria* was concerned with anaerobic hydrolysis fermentation and secretion of extracellular enzymes (Zhou et al., 2015). Besides, the relative abundances of *Clostridia* and *Bacteroidia* responsible for organic matters hydrolysis and fermentation (Xie et al., 2014, 2016), were 0.9% and 0.02% in biofilm, and higher than those in AS (0.4% and 0.003%). *ML635J-21* and *SC3-20* were detected in biofilm while were not detectable in AS. These results

indicated that PE carriers could enhance the plentifulness of microbial community and create a micro-ecological environment to enrich bacteria responsible for biological hydrolysis of organic matters.

Fig. 1c demonstrates the genus taxonomic level identification of the microbial communities. AS and biofilm had similar genera compositions but different relative abundances for specific genus. The genera *Anaerolineaceae_uncultured* (14.1–18.9%), *Caldilineaceae_uncultured* (5.3–11.2%), *Blastocatella* (3.4–5.3%), *Gemmatimonadaceae_uncultured* (3.6–6.0%), *Rhodocyclaceae_uncultured* (3.6–7.2%) and *Saprospiraceae_uncultured* (2.8–3.3%) were dominant in both AS and biofilm. *Anaerolineaceae*, a semi-syntrophic and fatty acid oxidizing bacteria, was considered to be responsible for carbohydrate couples degradation with hydrogenotrophic methanogens (Ding et al., 2016). *Blastocatella* of phylum *Acidobacteria*, the chemoheterotrophic bacteria correlated with hydrolysis, were also found to survive in the anaerobic side-stream reactor (Cheng et al., 2017). *Saprospiraceae_uncultured*, belonging to phylum *Bacteroidetes*, could be capable of hydrolyzing carbon polymers and preying of other bacteria (Cheng et al., 2018). Besides, some minor genera, including *Acidaminobacter*, *Fusibacter* and *Gelria*, were enriched in biofilm, with relative abundance of 0.10%, 0.09% and 0.02%, rather than AS with relative abundance of 0.003%, 0.003% and 0.01%. These genera were determined to be the protein- and carbohydrate-fermenting bacteria reported previously by other researchers (Chen and Chang, 2017). The genera *Ramlibacter*, only present in biofilm, was an anaerobic and or facultative anaerobic bacterium reported to slowly grow by the reduction of oxyanions, such as nitrite, nitrate, sulfate, and perchlorate (Zhou et al., 2014).

In summary, microorganisms accumulated on the surface of PE carriers were mainly originated from AS, while the population composition was changed (Ma et al., 2013b). Compared to those in AS, *Betaproteobacteria*, *Gammaproteobacteria*, *Sphingobacteria*, *Clostridia* and *Bacteroidia*, which could play an important role in biological hydrolysis of organic matters, were found to be the pioneers involved in the surface colonization on the PE carriers.

3.2. Kinetics of organic substances hydrolysis

Fig. 2 shows effects of carrier filling ratio, temperature and limited aeration on COD removal in the hydrolysis process. Variations of COD concentration in the supernatant with time in Fig. 2 were fitted by Eq. (2), and the results are illustrated in Table 3. Higher coefficients of determination in the fit curve mean higher credibility to the first-order kinetics. Hydrolysis not only increased COD removal, but also intensified biodegradability of organic substances in wastewater. Fig. 3 demonstrates effects of carrier filling ratio, temperature and limited aeration on BOD_5/COD ratio in the hydrolysis process.

3.2.1. Effects of carriers filling ratio

As shown in Fig. 2a, there was an obvious downtrend in the COD concentration with the increase of the carrier filling ratio in hydrolysis process. At the end of the reaction (48 h), SCOD concentrations in the supernatant were 134.0, 105.3, and 94.2 mg/L at filling ratio of 0, 30% and 60%, respectively. Higher carrier filling ratio provided more biomass for hydrolysis, and resulted in better COD removal efficiency (Gu et al., 2014). The COD biodegradation process appeared to follow the first-order kinetic, and the biodegradation rate depended on the carrier filling ratio. With carrier filling ratio rising from 0 to 60% (Table 3), k increased 0.015 to 0.080 h^{-1}, while the inert fraction COD_I decreased from 119.5 to 94.0 mg COD/L. It could be explained that packing PE carriers increased the population and abundance of microorganisms and

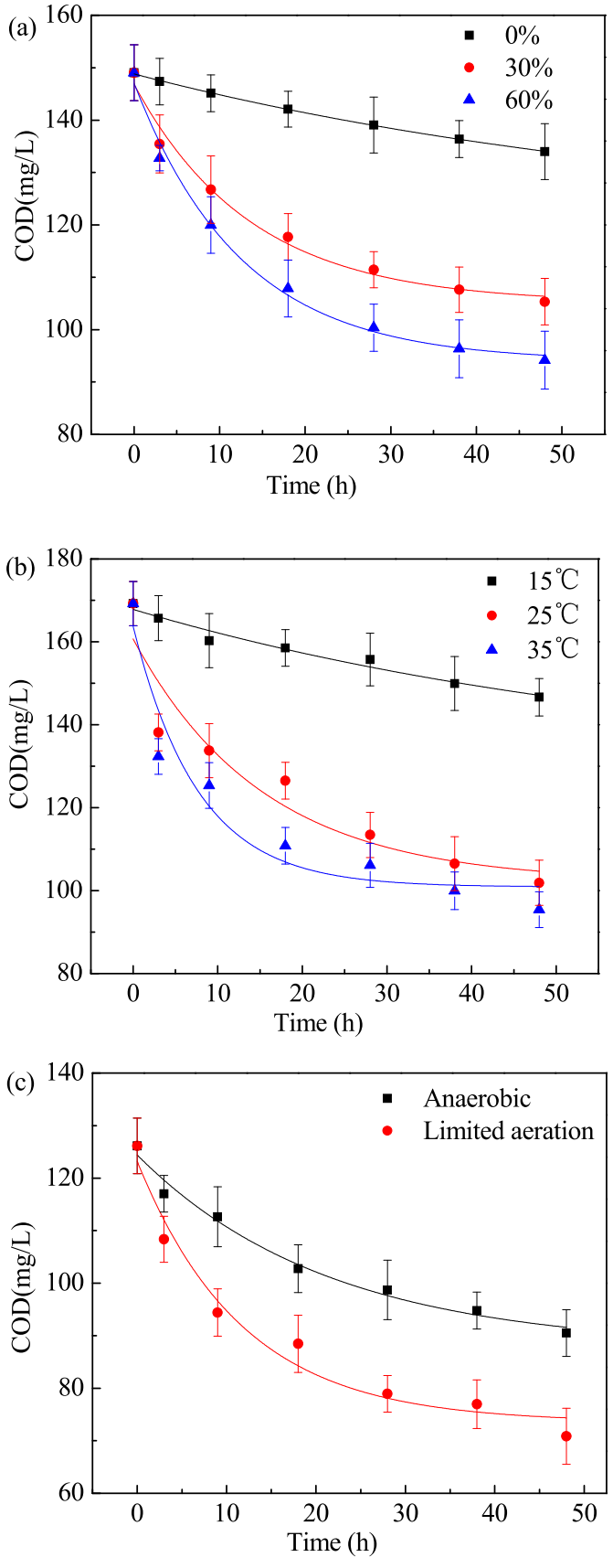


Fig. 2. Effects of (a) carrier filling ratio, (b) temperature, and (c) limited aeration on the changes of COD concentration during the hydrolysis process.

Table 3
Kinetic parameters for COD removal during hydrolysis fitted by first-order kinetic model.

Influential factor		Regression equation	R^2	k (h^{-1})	COD_0 (mg/L)	COD_I (mg/L)
Carrier filling ratio	0	$COD = 29.4\ e^{-0.015\ x} + 119.5$	0.993	0.015	29.4	119.5
	30%	$COD = 41.8\ e^{-0.073\ x} + 105.1$	0.981	0.073	41.8	105.1
	60%	$COD = 53.1\ e^{-0.080\ x} + 94.0$	0.990	0.080	53.1	94.0
Temperature	15 °C	$COD = 39.7\ e^{-0.015\ x} + 128.1$	0.956	0.015	39.7	128.1
	25 °C	$COD = 58.6\ e^{-0.065\ x} + 102.1$	0.872	0.065	58.6	102.1
	35 °C	$COD = 62.9\ e^{-0.130\ x} + 100.9$	0.908	0.130	62.9	100.9
Limited aeration	Anaerobic	$COD = 36.7\ e^{-0.047\ x} + 87.7$	0.979	0.047	36.7	87.7
	Limited aeration	$COD = 49.7\ e^{-0.085\ x} + 73.5$	0.961	0.085	49.7	73.5

enhanced the ability of organic matter degradation, so more refractory organics in the wastewater was degraded into small molecules.

At the beginning, the BOD_5/COD ratio in the supernatant of raw wastewater was only 0.11, and elevated to 0.20 and 0.29 after hydrolysis for 24 and 48 h without carriers, indicating that biological treatment process was inappropriate even after hydrolysis. However, the ratio significantly could augment to 0.42 and 0.54 after hydrolysis for 48 h with carriers filling ratio of 30% and 60% (Fig. 3a). The results indicated that packing carriers could improve the biodegradability of the industrial park wastewater, and increasing filling ratio further enhanced biodegradability of unbiodegradable organic pollutants and favored COD removal in the succedent biological process.

3.2.2. Effects of temperature

Temperature is an important factor affecting a large number of biological processes (Li et al., 2015), including hydrolysis (He et al., 2012). According to Fig. 2b, with the temperature increment from 15 °C to 25 °C and 35 °C, the COD removal efficiencies increased continuously and reached 13.3%, 39.8%, and 43.6% after 48 h hydrolysis, respectively. Obviously, higher temperature achieved relative higher COD removal efficiency by promoting metabolism rate to enhance COD removal (Li et al., 2015). Compared with carrier filling ratio, similar variation trends of k and COD_I with temperature could also be observed (Table 3). When the temperature was adjusted from 15 °C to 35 °C, k was significantly increased from 0.015 to 0.130 h^{-1}, while COD_I was reduced from 128.1 to 100.9 mg COD/L. It was shown that increasing the temperature enhanced enzymatic activity for hydrolysis, and the influence of temperature on k can be described by Arrhenius equation (Tian et al., 2019). The Ea of hydrolysis process with carriers for COD removal from industrial park wastewater was calculated to be 80.22 kJ/mol, which was slightly greater than the typical value for enzyme kinetics reaction (20–80 kJ/mol) (Zhang et al., 2016).

Temperature can greatly influence the thermodynamic equilibrium of biochemical reactions and microbial community in the biological hydrolysis process (Lin et al., 2016, 2017). For example, *Betaproteobacteria* and *Clostridia* showed evidently increased percentages with the increment in temperature (Chen and Chang, 2017), reflecting that the growth of the functional hydrolysis bacteria was elevated with rising temperature. The BOD_5/COD ratio obtained with time at different temperatures were given in Fig. 3b. The BOD_5/COD ratio increased to 0.27, 0.42 and 0.66 after 48 h hydrolysis with carriers at filling ratio of 30% at 15, 25 and 35 °C, respectively. The results indicated that increasing temperature benefited to enhancing biodegradability. It should be noted that, the biodegradability of organic substance in the industrial park wastewater was slightly improved under low temperature even after packing carriers, which would obstacle COD removal in winter. Heating and insulation measures are proposed to be employed for the hydrolysis unit if waste heat could be provided in the industrial park.

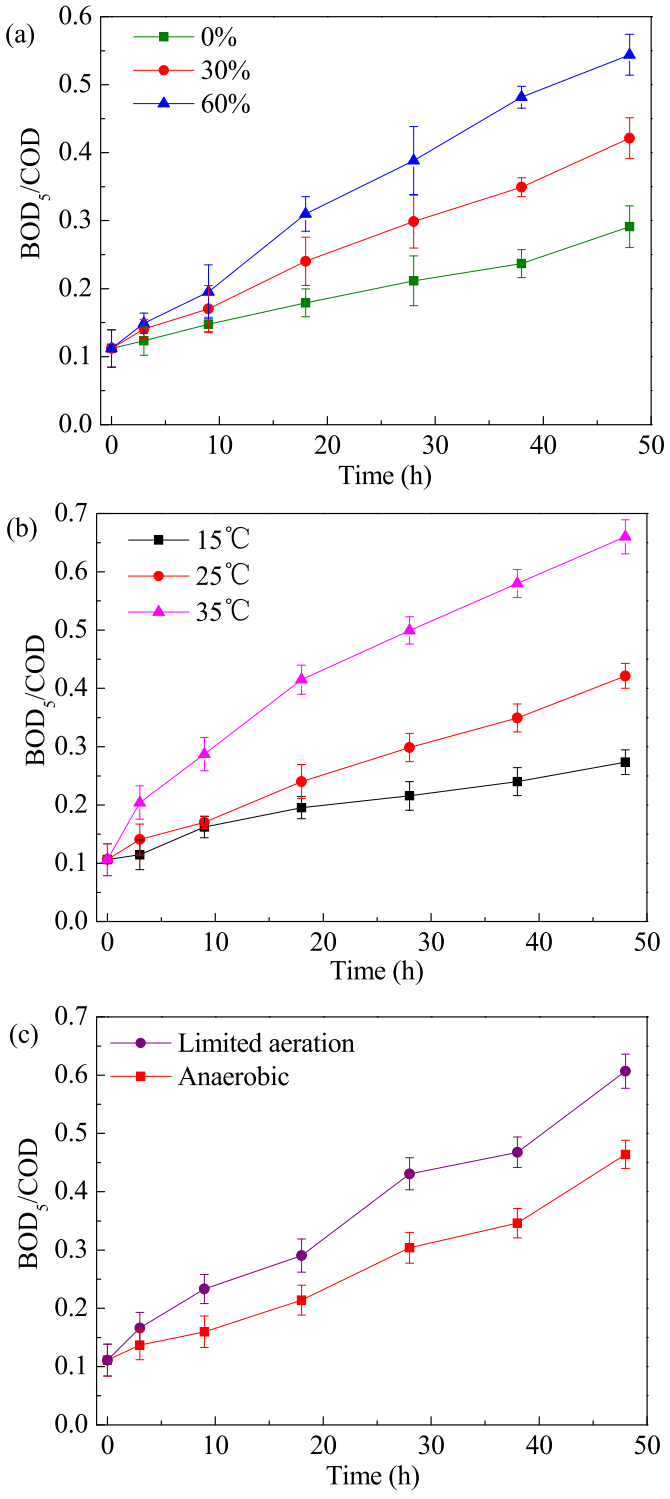


Fig. 3. Effects of (a) carrier filling ratio, (b) temperature, and (c) limited aeration on the changes of BOD_5/COD ratio during the hydrolysis process.

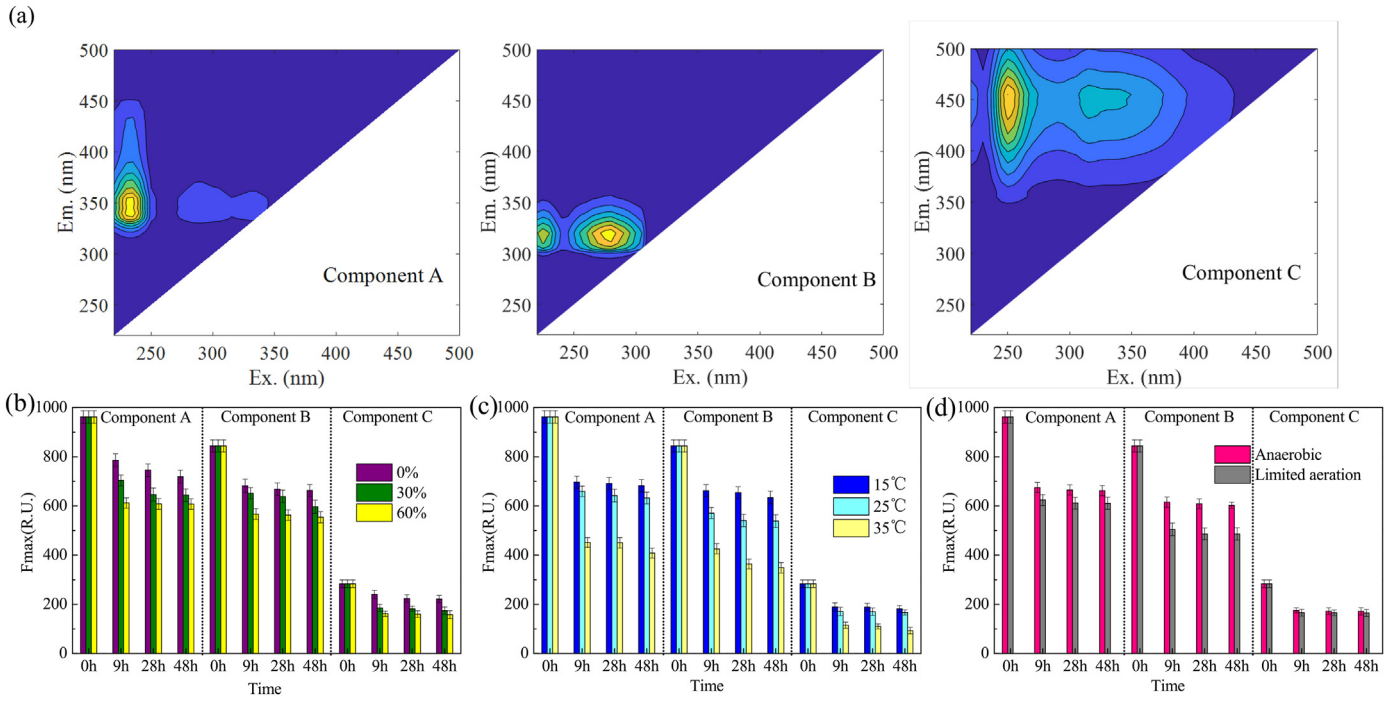


Fig. 4. Removal of (a) the three PARAFAC components at different (b) carrier filling ratio, (c) temperature, and (d) limited aeration during the hydrolysis process.

3.2.3. *Effects of limited aeration*

It can be perceived that limited aeration is an environmentally friendly, rapid and safe technology for the hydrolysis pretreatment of a mixed wastewater (Wu et al., 2015), including domestic wastewater, pharmaceutical industry wastewater, petroleum industrial wastewater, textile dye industry wastewater, etc. As illustrated in Fig. 2c, the COD removal rate in the limited aeration reactor (43.8%) was higher than that in the anaerobic reactor without aeration (28.3%). As for kinetics, limited aeration enhanced parameter k by 80.8%, and reduced COD_l from 87.7 mg COD/L to 73.5 mg COD/L. The results indicated that limited aeration accelerate hydrolysis of complex organic matters by stimulating biological activities of both anaerobic and aerobic organisms (Niu et al., 2016).

As shown in Fig. 3c, limited aeration also enhanced biodegradability as BOD_5/COD ratio rose from 0.46 to 0.61 after 48 h hydrolysis with carriers at filling ratio of 30%. Besides stimulating hydrolytic activity, limited aeration also prohibited the consumption of biodegradable BOD in the sulfate reduction process. *Illumina-MiSeq* sequencing results showed that the predominant microorganisms in biofilm of the limited aeration reactor and the anaerobic reactor exhibited a high homology (Fig. 1). According to Wu et al. (2015), low DO condition (0.2–0.3 mg/L) could weak or block SO_4^{2-} reduction to a certain degree in the hydrolysis acidification process, ultimately resulting in the less generation of H_2S and fewer consumption of BOD. That could improve the biodegradation and treatability for the hydrolysis process. It should be noted that (Fig. 1c) the total abundance of the sulfate-reducing bacteria *Desulfobacula*, *Desulfobulbus*, *Desulfocapsa*, *Desulfococcus*, *Desulfofustis*, *Desulfomicrobium*, *Desulforhabdus*, *Desulfovibrio*, *Desulfurispora*, and *Desulfuromonas*, which could reduce sulfate to hydrogen sulfide by using organic matters, were detected in biofilm of the limited aeration reactor (0.42%) and lower than that in the anaerobic reactor (0.71%). The relative abundances of *Anaerolineaceae* and *Blastocatella*, which could strengthen hydrolysis, also significantly increased in biofilm of the limited aeration reactor (21.1% and 4.5%) compared to that in the anaerobic reactor (14.1% and 3.4%).

3.3. *Variations of fluorescent organic substances during hydrolysis*

In this study, PARAFAC analysis was employed to characterize changes of fluorescence DOM during hydrolysis process of the wastewater (Fig. 4). Residual analysis gave evidence of three components being adequate for these data. Then, Component A, B and C, detected at the excitation/emission wavelengths (Ex/Em) of 235/350 (225, 280)/320 and 250/455 nm (Fig. 4a), were identified as aromatic protein-like (Chen et al., 2003; Zheng et al., 2019), tryptophan protein-like (e.g., aromatic amino acids) (An et al., 2017; Tang et al., 2010; Zhao et al., 2018), and fulvic acid-like substances with aromatic terrestrial structures and macromolecular weight (Chen et al., 2003; Hu et al., 2015; Zhao et al., 2018). As illustrated in Fig. 4b–d, the amount of refractory aromatic protein (Component A) and intractable fulvic acid (Component C) was the main fluorescent substances of DOM, and biodegradable tryptophan protein (Component B) occupied relatively small proportion. Interestingly, maximum fluorescent intensity (F_{max}) values of Component A, B and C all decreased with time under different carrier filling ratios, temperatures and limited aeration, which had an alike variation tendency to the COD (Fig. 2).

F_{max} of the three components decreased with increasing carrier filling ratio from 0 to 60%. For example, at the end of the reaction (48 h), F_{max} of Component A in DOM of the reactor decreased by 242.6, 317.7, and 354.2, while that of Component C decreased by 62.6, 109.7, and 126.1 following the carrier filling ratio increasing from 0 to 60%. F_{max} of Component A, B and C decreased as temperature increased from 15 °C to 35 °C, and reduced when using reactor with limited aeration. The refractory aromatic protein and intractable fulvic acid were non-biodegradable (Zheng et al., 2019), which would increase the difficulty of biological treatment of the industrial park wastewater. These findings are very encouraging, showing that higher packing carriers, higher temperature and

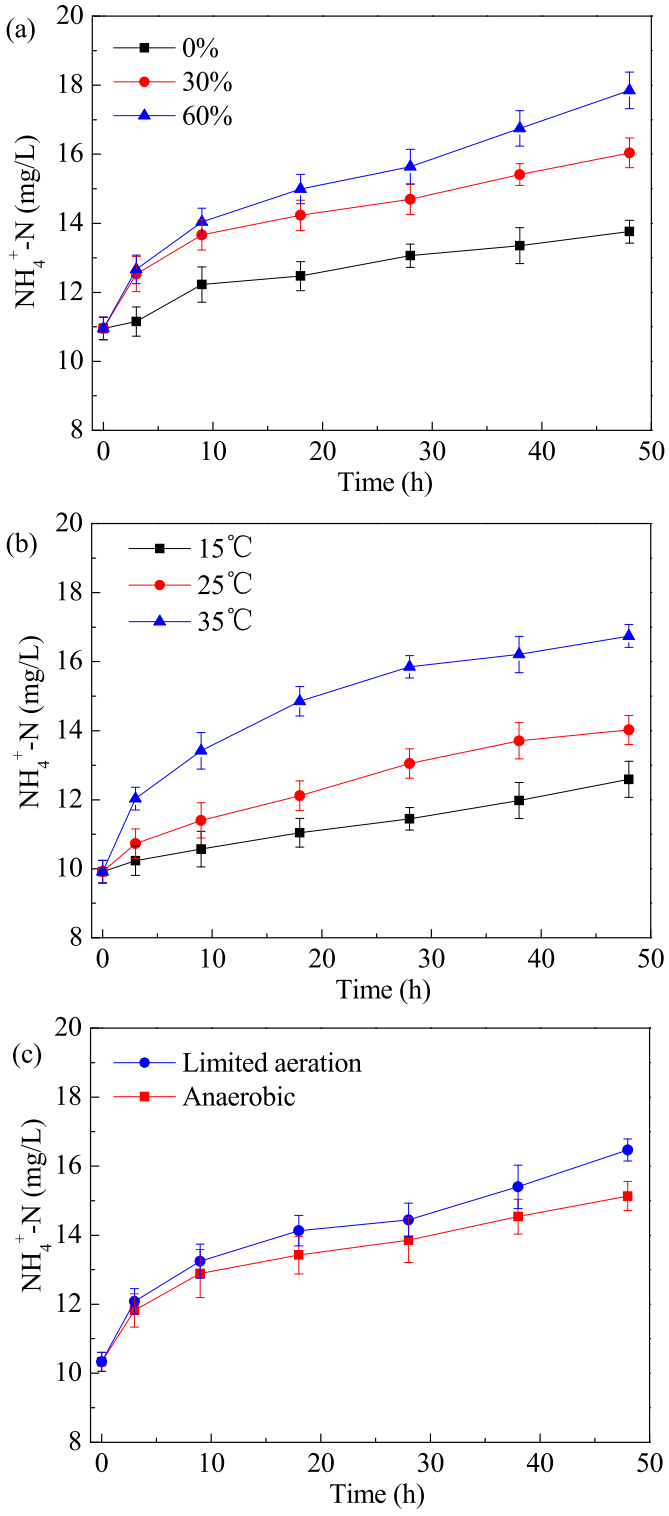


Fig. 5. Effects of (a) carrier filling ratio, (b) temperature, and (c) limited aeration on the change of NH_4^+-N concentration during the hydrolysis process.

limited aeration had the capacity to strengthen the proceeding of non-biodegradable substrate removal.

3.4. NH_4^+-N release during hydrolysis

Fig. 5 illustrates effects of the three influential factors on the change of NH_4^+-N concentration during the hydrolysis process. In the batch tests, the released NH_4^+-N concentrations were significantly correlated with COD decrement with correlation coefficients of 0.967, 0.901 and 0.947 at 0.01 level for Test 1, Test 2 and Test 3 (Table 1). The results confirmed that the release NH_4^+-N was transformed from the nitrogen-containing organic matters through decomposition (Bi et al., 2014; Zhang et al., 2017) in accompany with the progress of the hydrolysis reaction (Lin et al., 2013), and could also be employed to indicate hydrolysis efficiency. Organic nitrogen exists in wastewater containing compounds such as partially metabolized protein fragments, humic substances, urea, amino acids, nucleic acids, low-molecular weight amines, chelating agents, and et al. (Pehlivanoglu-Mantas and Sedlak, 2008; Zang et al., 2000). However, some nitrogen-containing organic matters are difficult to convert into NH_4^+-N during hydrolysis process, or to be degraded directly by heterotrophic bacteria in the subsequent process (e.g., aeration tank), resulting that TN removal was deficient to fulfill the discharge standard in some industrial park wastewater treatment plants (Pehlivanoglu-Mantas and Sedlak, 2008).

Compared to influent wastewater, conventional hydrolysis only released the concentration of NH_4^+-N by 2.8 mg/L after 48 h. As shown in Fig. 5a, the data increased to 5.1 and 8.9 mg/L when the carrier filling ratio was raised to 30% and 60%. The concentrations of released NH_4^+-N were 2.7, 4.1 and 6.8 mg/L after 48 h hydrolysis with 30% carriers at 15, 25 and 35 °C (Fig. 5b). It was found that higher carrier filling ratio or temperature could bring higher NH_4^+-N production, showing that adding PE carriers or increasing temperature was conducive to the degradation of nitrogen-containing organic matter. As shown in Fig. 5c, introducing limited oxygen also increased NH_4^+-N release by 15.1% after 48 h hydrolysis although it might resulted in partial oxidation of NH_4^+-N by nitrobacteria (Zhang et al., 2017). The total relative abundance of *Nitrospira* and *Nitrosomonas*, accounted for 0.61% of the total microbes in biofilm of the anaerobic reactor, and then increased to 0.97% in that of the limited aeration reactor (Fig. 1c). The results suggested that the actual release of NH_4^+-N in the limited aeration reactor was higher than the observed data in Fig. 5c.

The results obtained from this paper implied that the hydrolysis system with carrier filling and limited aeration provided an effective biological pre-treatment method for the chemical industrial park WWTP to reduce the organic loading and improve the biodegradability at normal temperature. For the functional bacteria in charge of hydrolysis, carriers provide a pleasant enough place to grow up (Zheng et al., 2019). Increasing the temperature strengthened the activity of enzymes and the metabolic capacity of microorganisms, but it is not economical to raise the temperature except waste heat is provided in adjacent region. Limited aeration could prohibit sulfate reduction to a certain degree in hydrolysis process and save BOD consumption (Wu et al., 2015). The COD removal efficiency and the average BOD_5/COD of the wastewater were both improved under the limited aeration condition at normal temperature (Figs. 2c and 3c). Packing carriers and limited aeration should be an appropriate and suitable method for upgrading the performance of hydrolysis system in the chemical industrial park WWTP. On the other hand, investigating pollutants degradation and bacterial evolution by individual fractions prepared in artificial wastewater or separated from real wastewater is a very important

research area for industrial park wastewater with complicated fractions, and more clear mechanisms on interactions between individual fractions and microbial community is beneficial to the development of novel technologies and enhancement strategies for wastewater treatment.

4. Conclusions

Packing carriers in hydrolysis tank intensified the plentifulness of microbial community and enriched hydrolytic bacteria (*Betaproteobacteria*, *Gammaproteobacteria*, *Sphingobacteria*, *Clostridia*, *Bacteroidia*). Rising carriers filling ratio increased first-order kinetic constant of hydrolysis from 0.015 to 0.080 h^{-1}, and enhanced biodegradability of COD in the industrial park wastewater. Rising temperature stimulated enzymatic activity for hydrolysis with *Ea* of 80.22 kJ/mol. Limited aeration further enriched hydrolytic bacteria, inhibited sulfate-reducing bacteria and intensified biodegradability of organic substances. Variations of fluorescent organic substances and NH_4^+-N release during hydrolysis confirmed that hydrolysis was strengthened by packing carriers, increasing filling ratio, enhancing temperature and introducing limited aeration. Packing carriers and limited aeration were techno-economic methods for upgrading the performance of hydrolysis system in the chemical industrial park WWTP by enriching hydrolytic bacteria and stimulating their activities.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

CRediT authorship contribution statement

Ying An: Conceptualization, Data curation, Writing - original draft. **Mengyu Wang:** Writing - original draft. **Zhen Zhou:** Methodology, Writing - review & editing, Supervision. **Xiao Sun:** Resources. **Cheng Cheng:** Formal analysis. **Songmei Zhu:** Investigation. **Luochun Wang:** Supervision. **Zhichao Wu:** Resources.

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References

An, Y., Zhou, Z., Yao, J., Niu, T.H., Qiu, Z., Ruan, D.N., Wei, H.J., 2017. Sludge reduction and microbial community structure in an anaerobic/anoxic/oxic process coupled with potassium ferrate disintegration. Bioresour. Technol. 245 (Part A), 954–961.

Bi, W., Li, Y.Y., Hu, Y.Y., 2014. Recovery of phosphorus and nitrogen from alkaline hydrolysis supernatant of excess sludge by magnesium ammonium phosphate. Bioresour. Technol. 166, 1–8.

Botheju, D., Bakke, R., 2011. Oxygen effects in anaerobic digestion - a review. Open Waste Manag. J. 4, 1–19.

Chen, H., Chang, S., 2017. Impact of temperatures on microbial community structures of sewage sludge biological hydrolysis. Bioresour. Technol. 245, 502–510.

Chen, W., Westerhoff, P., Leenheer, J.A., Booksh, K., 2003. Fluorescence excitation-emission matrix regional integration to quantify spectra for dissolved organic matter. Environ. Sci. Technol. 37 (24), 5701–5710.

Cheng, C., Zhou, Z., Niu, T.H., An, Y., Shen, X.L., Pan, W., Chen, Z.H., Liu, J., 2017. Effects of side-stream ratio on sludge reduction and microbial structures of anaerobic side-stream reactor coupled membrane bioreactors. Bioresour. Technol. 234, 380–388.

Cheng, C., Zhou, Z., Qiu, Z., Yang, J.Y., Wu, W., Pang, H.J., 2018. Enhancement of sludge reduction by ultrasonic pretreatment and packing carriers in the anaerobic side-stream reactor: performance, sludge characteristics and microbial community structure. Bioresour. Technol. 249 (Suppl. C), 298–306.

Chinese NEPA, 2012. Water and Wastewater Monitoring Methods, fourth ed. Chinese Environmental Science Publishing House, Beijing, China.

Choi, J., Kim, E.S., Ahn, Y., 2017. Microbial community analysis of bulk sludge/cake layers and biofouling-causing microbial consortia in a full-scale aerobic membrane bioreactor. Bioresour. Technol. 227 (Suppl. C), 133–141.

Ding, P.Y., Chu, L.B., Wang, J.L., 2016. Biological treatment of actual petrochemical wastewater using anaerobic/anoxic/oxic process and the microbial diversity analysis. Appl. Microbiol. Biotechnol. 100 (23), 10193–10202.

Gu, Q.Y., Sun, T.C., Wu, G., Li, M.Y., Qiu, W., 2014. Influence of carrier filling ratio on the performance of moving bed biofilm reactor in treating coking wastewater. Bioresour. Technol. 166, 72–78.

Han, X.M., Wang, Z.W., Ma, J.X., Zhu, C.W., Li, Y.X., Wu, Z.C., 2015. Membrane bioreactors fed with different COD/N ratio wastewater: impacts on microbial community, microbial products, and membrane fouling. Environ. Sci. Pollut. Control Ser. 22 (15), 11436–11445.

He, M.N., Sun, Y.B., Zou, D.X., Yuan, H.R., Zhu, B.N., Li, X.J., Pang, Y.Z., 2012. Influence of temperature on hydrolysis acidification of food waste. Procedia Environmental Sciences 16, 85–94.

Hu, D.L., Zhou, Z., Shen, X.L., Wei, H.J., Jiang, L.M., Lv, Y., 2015. Effects of alkalinity on membrane bioreactors for reject water treatment: performance improvement, fouling mitigation and microbial structures. Bioresour. Technol. 197, 217–226.

Hu, W.Q., Guo, Y., Tian, J.P., Chen, L.J., 2019a. Eco-efficiency of centralized wastewater treatment plants in industrial parks: a slack-based data envelopment analysis. Resour. Conserv. Recycl. 141, 176–186.

Hu, W.Q., Tian, J.P., Zang, N., Gao, Y., Chen, L.J., 2019b. Study of the development and performance of centralized wastewater treatment plants in Chinese industrial parks. J. Clean. Prod. 214, 939–951.

Huang, C.K., Shi, Y.J., Xue, J.K., Zhang, Y.Y., Gamal El-Din, M., Liu, Y., 2017. Comparison of biomass from integrated fixed-film activated sludge (IFAS), moving bed biofilm reactor (MBBR) and membrane bioreactor (MBR) treating recalcitrant organics: importance of attached biomass. J. Hazard Mater. 326, 120–129.

Kim, S., Seol, E., Oh, Y.K., Wang, G.Y., Park, S., 2009. Hydrogen production and metabolic flux analysis of metabolically engineered Escherichia coli strains. Int. J. Hydrogen Energy 34 (17), 7417–7427.

Kong, Z., Li, L., Xue, Y., Yang, M., Li, Y.Y., 2019. Challenges and prospects for the anaerobic treatment of chemical-industrial organic wastewater: a review. J. Clean. Prod. 231, 913–927.

Kumar, M., Ou, Y.L., Lin, J.G., 2010. Co-composting of green waste and food waste at low C/N ratio. Waste Manag. 30 (4), 602–609.

Li, C., Zhang, Z., Li, Y., Cao, J.S., 2015. Study on dyeing wastewater treatment at high temperature by MBBR and the thermotolerant mechanism based on its microbial analysis. Process Biochem. 50 (11), 1934–1941.

Lin, L., Wan, C.L., Liu, X., Lee, D.J., Lei, Z.F., Zhang, Y., Tay, J.H., 2013. Effect of initial pH on mesophilic hydrolysis and acidification of swine manure. Bioresour. Technol. 136, 302–308.

Lin, Q., De Vrieze, J., He, G., Li, X., Li, J., 2016. Temperature regulates methane production through the function centralization of microbial community in anaerobic digestion. Bioresour. Technol. 216, 150–158.

Lin, Q., De Vrieze, J., Li, C., Li, J., Li, J., Yao, M., Hedenec, P., Li, H., Li, T., Rui, J., Frouz, J., Li, X., 2017. Temperature regulates deterministic processes and the succession of microbial interactions in anaerobic digestion process. Water Res. 123, 134–143.

Liu, H.B., Wang, H.N., Zhou, X., Fan, J.L., Liu, Y.F., Yang, Y.K., 2019. A comprehensive index for evaluating and enhancing effective wastewater treatment in two industrial parks in China. J. Clean. Prod. 230, 854–861.

Ma, J.X., Wang, Z.W., Yang, Y., Mei, X.J., Wu, Z.C., 2013a. Correlating microbial community structure and composition with aeration intensity in submerged membrane bioreactors by 454 high-throughput pyrosequencing. Water Res. 47 (2), 859–869.

Ma, J.X., Wang, Z.W., Zou, X.X., Feng, J.J., Wu, Z.C., 2013b. Microbial communities in an anaerobic dynamic membrane bioreactor (AnDMBR) for municipal wastewater treatment: comparison of bulk sludge and cake layer. Process Biochem. 48 (3), 510–516.

Ma, K., Qin, Z., Zhao, Z.Q., Zhao, C.X., Liang, S.X., 2016. Toxicity evaluation of wastewater collected at different treatment stages from a pharmaceutical industrial park wastewater treatment plant. Chemosphere 158, 163–170.

Matar, G.K., Bagchi, S., Zhang, K., Oerther, D.B., Saikaly, P.E., 2017. Membrane biofilm communities in full-scale membrane bioreactors are not randomly assembled and consist of a core microbiome. Water Res. 123 (Suppl. C), 124–133.

Niu, T.H., Zhou, Z., Shen, X.L., Qiao, W.M., Jiang, L.M., Pan, W., Zhou, J.J., 2016. Effects of dissolved oxygen on performance and microbial community structure in a micro-aerobic hydrolysis sludge in situ reduction process. Water Res. 90, 369–377.

Pehlivanoglu-Mantas, E., Sedlak, D.L., 2008. Measurement of dissolved organic nitrogen forms in wastewater effluents: concentrations, size distribution and NDMA formation potential. Water Res. 42 (14), 3890–3898.

Tang, S.J., Wang, Z.W., Wu, Z.C., Zhou, Q., 2010. Role of dissolved organic matters (DOM) in membrane fouling of membrane bioreactors for municipal wastewater treatment. J. Hazard Mater. 178 (1–3), 377–384.

Tian, X.C., Zhou, Z., Xin, Y., Jiang, L.-M., Zhao, X.D., An, Y., 2019. A novel sulfate removal process by ettringite precipitation with aluminum recovery: kinetics and a pilot-scale study. J. Hazard Mater. 365, 572–580.

Wu, C.Y., Zhou, Y.X., Wang, P.C., Guo, S.J., 2015. Improving hydrolysis acidification by limited aeration in the pretreatment of petrochemical wastewater. Bioresour. Technol. 194, 256–262.
Wu, Q.L., Guo, W.Q., Zheng, H.S., Luo, H.C., Feng, X.C., Yin, R.L., Ren, N.Q., 2016. Enhancement of volatile fatty acid production by co-fermentation of food waste and excess sludge without pH control: the mechanism and microbial community analyses. Bioresour. Technol. 216, 653–660.
Xie, X.H., Liu, N., Yang, B., Yu, C.Z., Zhang, Q.Y., Zheng, X.L., Xu, L.Y., Li, R., Liu, J.S., 2016. Comparison of microbial community in hydrolysis acidification reactor depending on different structure dyes by Illumina MiSeq sequencing. Int. Biodeterior. Biodegrad. 111, 14–21.
Xie, Z.F., Wang, Z.W., Wang, Q.Y., Zhu, C.W., Wu, Z.C., 2014. An anaerobic dynamic membrane bioreactor (AnDMBR) for landfill leachate treatment: performance and microbial community identification. Bioresour. Technol. 161, 29–39.
Zang, X., van Heemst, J.D.H., Dria, K.J., Hatcher, P.G., 2000. Encapsulation of protein in humic acid from a histosol as an explanation for the occurrence of organic nitrogen in soil and sediment. Org. Geochem. 31 (7–8), 679–695.
Zhang, L., Gao, R., Naka, A., Hendrickx, T.L.G., Rijnaarts, H.H.M., Zeeman, G., 2016. Hydrolysis rate constants at 10–25 °C can be more than doubled by a short anaerobic pre-hydrolysis at 35 °C. Water Res. 104, 283–291.
Zhang, Z.G., Pan, S.L., Huang, F., Li, X., Shang, J.F., Lai, J., Liao, Y.T., 2017. Nitrogen and phosphorus removal by activated sludge process: a review. Mini-Reviews Org. Chem. 14 (2), 99–106.
Zhao, C., Wang, Z.H., Wang, C.Y., Li, X., Wang, C.C., 2018. Photocatalytic degradation of DOM in urban stormwater runoff with TiO2 nanoparticles under UV light irradiation: EEM-PARAFAC analysis and influence of co-existing inorganic ions. Environ. Pollut. 243, 177–188.
Zheng, Y., Cheng, C., Zhou, Z., Pang, H.J., Chen, L.Y., Jiang, L.M., 2019. Insight into the roles of packing carriers and ultrasonication in anaerobic side-stream reactor coupled membrane bioreactors: sludge reduction performance and mechanism. Water Res. 155, 310–319.
Zhou, Z., Qiao, W.M., Xing, C., An, Y., Shen, X.L., Ren, W.C., Jiang, L.M., Wang, L.C., 2015. Microbial community structure of anoxic-oxic-settling-anaerobic sludge reduction process revealed by 454-pyrosequencing. Chem. Eng. J. 266, 249–257.
Zhou, Z., Qiao, W.M., Xing, C., Shen, X.L., Hu, D.L., Wang, L.C., 2014. A micro-aerobic hydrolysis process for sludge in situ reduction: performance and microbial community structure. Bioresour. Technol. 173, 452–456.
Zhu, S.M., Zhou, Z., Jiang, H.T., Ye, J.F., Ren, J.M., Gu, L.Y., Wang, L.C., 2016. Advanced treatment of effluents from an industrial park wastewater treatment plant by ferrous ion activated persulfate oxidation process. Water Sci. Technol. 72 (2), 535–541.